

RSBU Project Team Organisation Transition Rules

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Hitachi Rail – Internal

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L1. G-TRN A08001 RSBU Project Team Organisation Transition Rules

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1 SCOPE

1.1 Purpose

The purpose of this document is to establish guidelines and describe critical roles supporting the Bidding and Project execution phases. During the transition period (see. §1.3 Effective Date), this document ensures consistency and compliance on how the Railway Systems Business Unit (hereinafter RSBU or the Company) allocates responsibilities and functional relationships for key roles within the teams operating in Bidding and Project execution phases. It also defines the relationship between the functional and team organisations.

1.2 Applicability

Refer to the table below for the applicability of this document.

Legal Entities	All RSBU Legal Entities
Line of Business (LoB)	Line of Business (LoB) Signalling and Rail Solutions (SRS) and Line of Business Vehicles (VHC), regarding a single LoB bid/project
Project/Bid Type/Criteria	Customer Projects in Bidding and Project Execution
Additional Criteria and Notes	This document does not apply to current Hitachi Rail Japan (HRJ) business tenders, bids, and projects because these are managed according to local specific roles. This document does not apply to internal projects (e.g. R&D, Development and Innovation Projects). For bids currently in progress and project currently in execution refer to §5

1.3 Effective Date

This present document maps the guideline during the transition period. It is valid as soon as it published and until superseded by the publication of a fully harmonised Project Team Organisation MNL.

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2 TERMS

2.1 Acronyms and Abbreviations

BEP	BIM Execution Plan
BIM	Building Information Modelling
BM	Bid Manager
CCB	Change Control Boards
CI	Continuous Improvement
DDQ	Design, Develop and Qualify
EAC	Estimate At Completion
FAI	First Article Inspection
FAS&T	Finance, Accounting, Services & Treasury
FAT	Factory Acceptance Test
FRACAS	Failure Reporting, Analysis, and Corrective Action System
FTE	Full Time Equivalent
HRJ	Hitachi Rail Japan
HSE	Health, Safety, and Environment
IEC	International Electrotechnical Commission
JV	Joint Venture
KPI	Key Performance Indicator
LCC	Life Cycle Costs
LoB	Line of Business
MPR	Monthly Project Review
MWPM	Main Work Package Manager
NPV	Net Present Value
O&M	Operation and Maintenance
PE	Project Engineer
P&L	Profit and Loss
PJC	Project Controller
PJP	Project Planner

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PM	Project Manager
POM	Project Operations Manager
PQM	Project Quality Manager
PQMP	Project Quality Management Plan
PPP	Public-Private Partnership
PTO	Project Team Organisation
QA	Quality Assurance
QPR	Quarterly Project Review
RAM	Reliability, Availability and Maintainability
RAMS	Reliability, Availability, Maintainability, and Safety
RFI	Request for Information
RFQ	Request for Quotation
S&M	Service and Maintenance
SCPM	Supply Chain and Procurement Manager
SLA	Service Level Agreement
SRAC	Safety Related Application Conditions
SRS	Signalling and Rail Solutions
SoW	Scope of Work
T&Cs	Terms and Conditions
V&V	Verification and Validation
VHC	Vehicles
WOB	Workload Optimisation and Balancing
WP	Work Package
WPL	Work Package Leader
WPM	Work Package Manager

2.2 Definitions

Commissioning

Phase before handover to a Customer, in which a product is final tested under operational conditions to verify it functions according to its specifications (as per ISO 22163:2023).

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Customer	Person or organisation that could or does receive a product or a service that is intended for or required by this person or organisation (as per ISO 9000:2015).
Key Initiatives (not applicable to VHC LoB)	Key initiatives represent Turnkey/PPP Projects (with or without Partnerships with other Companies) with relevant System Integration and Interface Management content. Other major projects can be defined as Key Initiative time by time in agreement with the relevant Line of Business
Maintenance	Combination of all technical and management actions intended to retain an item in, or restore it to a state in which it can perform as required (as per ISO 22163:2023).
Organisation	Person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its objectives (as per ISO 9000:2015, ISO 14001:2015, ISO 45001:2018).
Project	Unique process, consisting of a set of coordinated and controlled activities with start and finish dates, undertaken to achieve an objective conforming to specific requirements, including the constraints of time, cost and resources (as per ISO 9000:2015).
Project management	Planning, organizing, monitoring, controlling and reporting of all aspects of a project, and the motivation of all those involved in it to achieve the project objectives (as per ISO 9000:2015).
Provider	Organisation that provides a product or a service (as per ISO 9000:2015).
Supply Chain	System of organisations, people, activities, information and resources involved in transforming materials and knowledge in a product or a service for the customer (as per ISO 22163:2023)
Technical Authority	The Technical Authority is a senior technical individual who analyses, challenges, validates and approves the technical solution and the technical delivery process proposed for Bids or Projects, ensuring technical risks have been minimized with proper mitigation plans through the critical phases from bid to completion of project execution

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3 REFERENCES

3.1 Hitachi Corporate Reference Documents

Unless otherwise noted, references apply to the latest approved revision.

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3.2 Management Systems Reference Documents

Unless otherwise noted, references apply to the latest approved revision.

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3.3 Applicable Standards and Regulations

ISO 9000:2015	Quality management systems - Fundamentals and Vocabulary
ISO 14001:2015	Environmental management systems – Requirements with guidance for use
ISO 45001:2018	Occupational health and safety management systems – Requirements with guidance for use
ISO 22163:2023	Railway applications — Railway quality management system — ISO 9001:2015 and specific requirements for application in the railway sector

4 TRANSITION RULES

4.1 Organisational Principles

4.1.1 Organisational Key Dimension

For the purpose of Bidding and Project execution, RSBU operates through a two-dimensional matrix structure, combining:

- the functional organisation: grouping specific organisational functions with job specific accountabilities and competences;
- the bidding team organisation or the project team organisation: aimed at ensuring effective and efficient delivery of Bidding and Project execution phases.

4.1.2 Functional Organisation

The functional organisation is mapped in RSBU Organisational Chart (published on COSMO/COSMIC).

The functional organisation ensures:

- standardisation of processes, products (where applicable) and competences by ensuring the use of common procedures and tools;
- resources staffing on bids and projects;
- workload and capacity planning;
- development of the resource's competences;
- capitalisation and sharing of the know-how.

The details of the key accountabilities assigned to the Company's organisational functions are described in Roles and Mandates/functional job descriptions.

The operative bridge between the functional organisation and the business delivery is the Project Team Organisation (PTO), established to achieve specific bid or project objectives and in charge of the bidding or project activities.

4.1.3 Bidding Team Operating Model and Project Execution Team Operating Model

Bidding Teams and Project Teams are the key pillars of RSBU delivery processes.

Collaboration, cross-functional resourcing and a process approach are the key success factors in the deployment of RSBU's business strategy, as no individual or single organisational function holds all the required levers needed to achieve the business objectives.

Integrated project teams effectively staffed with required skills, competences and level of resourcing are the key drivers needed to achieve Company goals.

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Each Bidding or Project Team is led by a single accountable leader, respectively the Bid Manager (BM) during the bidding phase and the Project Manager (PM) during the project execution phase, who holds overall accountability.

The BM/PM is ultimately accountable for the staffing of the overall team organisation, with their direct reporting team members (such as PE) responsible for the staffing of their subordinate project structure. Based on Scope of Work (SoW) and requirements, the BM/PM is accountable for providing staffing requirements to the functional organisation. The functional organisation is accountable for allocating the most appropriate profiles, to be agreed with the BM/PM.

Depending on the Scope of Work (SoW) and requirements highlighted by the Bid Manager or Project Manager and communicated to the relevant functional organisation, the direct reporting Team Members are proposed by the functional managers from the appropriate functions within the Organisation and agreed with the Bid/Project Manager. The functional organisation analyses the Bid/Project requirements with the Bid or Project Manager and allocates the most appropriate profiles, ensuring correct allocation of skills and workload.

Whilst Team Members are identified from their relevant functions, they are managed by the Bid Manager or Project Manager (potentially via other managers within the Bid or Project), and are accountable for the delivery of their SoW, including all associated responsibilities in term of scope, schedule, cost, quality and risks relevant to Bidding or Project objectives.

Team Members are required to conduct their activities in accordance with relevant RSBU processes.

4.1.4 Work Breakdown Structure and Accountability Model

The Bid or Project SoW is structured through a Work Breakdown Structure (WBS), a hierarchical decomposition of a project into smaller, manageable components called work packages. The levels of breakdown may vary on the nature and complexity of the Bid or Project, with the lowest level defined as a Work Package (representing the level at which scope, schedule, cost, deliverables and performance can be planned, estimated, monitored, controlled, and delivered).

The structure of the Bidding or Project team will reflect the structure of the WBS, with four levels of role envisioned:

1. **Bid/Project Manager:** Overall accountable for the Bid or Project delivery
2. **Main Work Package Manager (MWPM):** Level 1 reporting to the Bid or Project Manager, accountable for the first level of breakdown of the WBS (often related to an overall function such as Engineering)
3. **Work Package Manager (WPM):** Level 2 or lower to the Bid or Project Manager, reporting to a MWPM or another WPM with accountability for a section of the WBS and the management of subordinate work packages

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4. **Work Package Leader (WPL):** Level 2 or lower to the Bid or Project Manager, reporting to a MWPM or WPM and accountable for the delivery of a Work Package, the lowest level of the WBS.

Depending on the complexity of the WBS, this hierarchy may collapse into fewer layers.

The current PTO document covers the Bid/Project Manager, and the Main Work Package Manager roles. Following a structured approach, it will continue to be further regulated with additional standard WPM and WPL roles in future document publications, driven by the significance of the roles and the detail of the WBS.

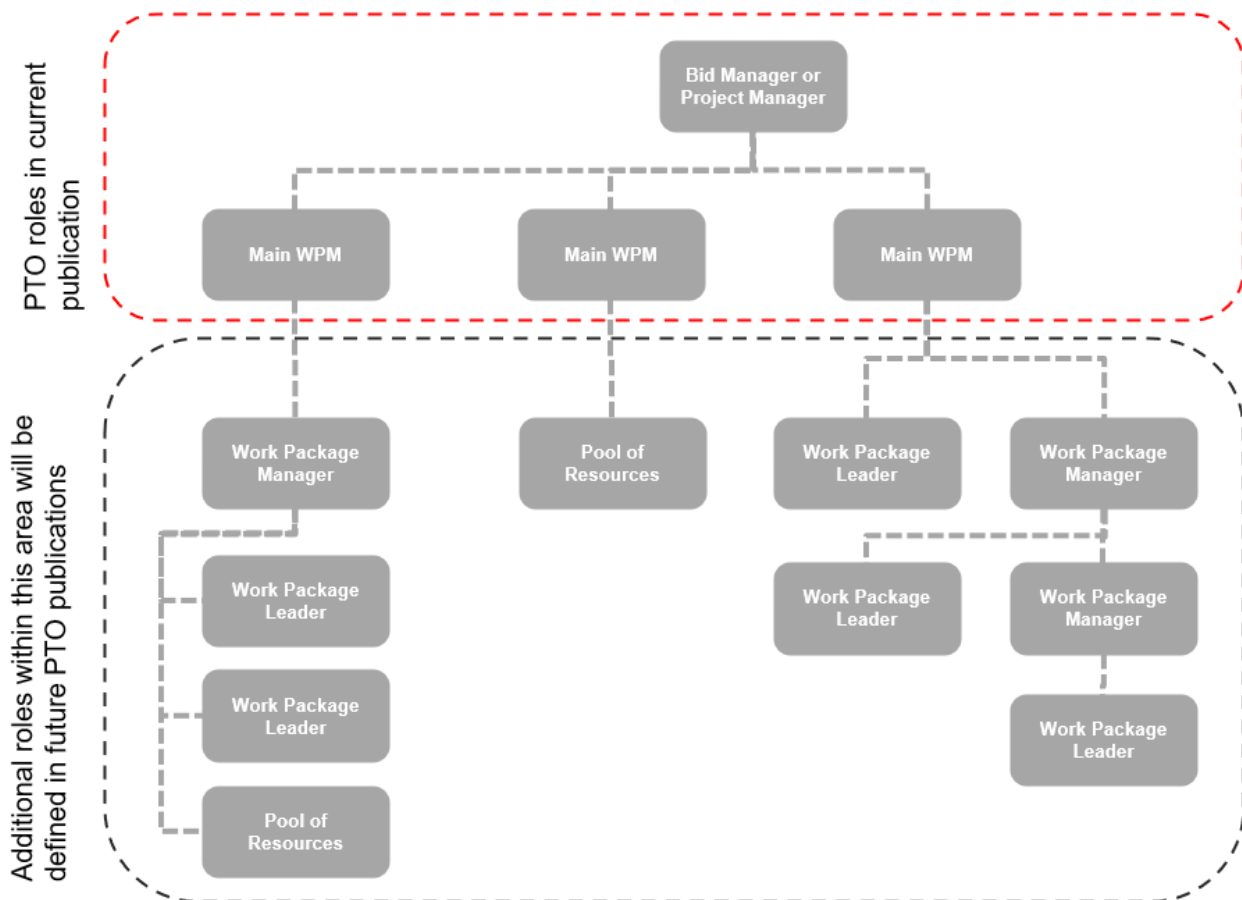


Figure 1: Example of team structure

Each MWPM/WPM/WPL is accountable for defining and managing the required WP activities to comply with the requirements, including safety, quality, cost estimation and monitoring, as well as the scheduling and performance control of an assigned pool of resources, aligned to the bid or project team common schedule. In case of issues, they promote an appropriate escalation.

The functional organisation to which the MWPM, WPM, and WPLs belong are accountable for:

- providing competent resources and standard methods and tools to the teams;

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- ensuring the implementation across bids and projects of standard and efficient solutions (procedural and technical), in compliance with the strategic guidelines;
- ensuring adherence to quality, RSBU processes and schedule targets for the deliverables produced by resources belonging to their organisation, and assisting their team members, as necessary, in identifying solutions to eliminate or reduce deviations and to improve performance;
- in addition, the managers of the functional organisations are responsible for approving the deliverables produced within their assigned scope in terms of quality and process, whether submitted directly to the Bid or Project Manager or through the relevant Main Work Package Manager;
- in conclusion, managers of the functional organisation remain accountable for the technical integrity, quality and adequacy of the deliverables produced within their scope throughout the bid or project lifecycle, including at key milestones and critical events where functional expertise and oversight are required, in co-accountability with their respective MWPMs, WPMs and WPLs;
- in the LoB Vehicles, managers of the functional organisation, with their respective MWPMs, WPMs, and WPLs, are additionally co-accountable on costs.

4.1.5 Issue Resolution and Escalation Principles

The Bid or Project Manager and the MWPMs, WPMs, and WPLs shall endeavour to resolve bidding or project execution issues internally before escalating them to the relevant functional organisation. Issues shall be escalated to the relevant functional organisation, as appropriate, when the required know-how is not available at Bidding or Project execution level, or when there is a reasonable risk of the team being unable to maintain control of delivery in line with targets.

In order to ensure seamless transition between the bidding and the project execution phase, the team should pursue continuity of individuals allocated to key roles (where possible). Additionally, Bidding team members should strongly support the Project Team during the initial phases of Project execution (PJM-01 Project Launch and PJM-02 Project Planning) in alignment with the relevant functional organisation.

Where alignment between the functional and project organisation (regarding staffing or tailoring) cannot be achieved in due time, the matter shall be escalated to the appropriate management level for resolution.

4.1.6 Team Tailoring and Governance

The Team Organisations described in this document represent the standard composition of the Bid or Project team and shall apply as the reference model.

It is the accountability of the Bid/Project Manager to ensure the alignment of the Bid/Project Team Organisation with the standard roles described in this document.

Only where formally justified by specific characteristics, contract requirements, or requirements of the Bid, Project, or Business, the Bid or Project Manager may tailor

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the Bid or Project Team organisation composition based on effectiveness and efficiency criteria in alignment with the functional organisation.

4.2 Bidding Team

4.2.1 Bidding Team – Standard Principles

The Bidding Team is the inter-functional organisational structure accountable for developing and managing the bid, under the leadership of the Bid Manager, until its closing.

A Bidding Team is established to pursue new orders whenever the RSBU takes part in a commercial opportunity, such as public tenders, unsolicited proposals, or direct negotiations, following a 'Go' decision and requiring a full bidding process.

Bidding Team Members are staffed by the relevant functional organisation based on the scope of work and the specific tender requirements and competences requested by the Bid Manager (BM). The composition of the Bidding Team shall be defined and aligned with the structure outlined in this document and may be tailored—where justified by bid characteristics, customer requirements or specific requirements—in alignment with the functional organisation.

Although the Bidding Team is expected to be staffed in accordance with this document, due to the variety in bid size and complexity, the following principles may be taken in consideration for tailoring:

- A Bidding Team role does not necessarily correspond to 1 Full Time Equivalent (FTE). Based on workload and complexity, the Bid Manager and the relevant functional organisation determine the appropriate FTE allocation
- In certain cases (e.g., small and less complex bids with highly experienced professionals), one individual may cover more than one role (either fully or partially), ensuring appropriate reporting within the functional organisation.

As key principles:

- A single Capture Leader is appointed for each Opportunity and is accountable for its commercial and strategic success, working closely with the Bid Manager throughout bid preparation.
- A single Bid Manager is appointed for each bid and leads the Bidding Team across all business segments, regions and legal entities, holding full end-to-end accountability for the bid, ensuring integrated technical, commercial, financial and governance robustness.
- A single Project Engineer acts as the sole technical lead on the bid solution, ensuring end-to-end technical coherence, integrity and feasibility.
- The BM and PE can be supported by Level-2 Bid Team roles in line with the WBS (e.g. WPM, WPL), subordinate within the bid hierarchy.

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- All operational Bid Team Members are accountable for executing their assigned scope within the Bid Team, while remaining functionally connected to their respective functional line and operationally managed by the Bid Manager.

Each Bidding Team Member is required to:

- perform the assigned activities on budget, quality and on time, and in line with Safety >> Quality > Delivery > Cost (S>>Q>D>C) approach
- analyse the bidding requirements in order to identify and structure the required Work Packages in accordance with the project WBS
- provide the Bid Manager with an estimate of hours, costs and requirements related to the execution of the activities within the assigned SoW/WPs
- provide the Project Manager with an evaluation of the risks related to the assigned SoW/WPs (as applicable)
- provide the Bid Manager with the related work schedule for the project delivery (where applicable)
- issue documentation as per the Deliverables List provided by the Bid Manager (where applicable)
- contribute to risk and opportunity analysis
- once the project is awarded, and under the coordination of the Bid Manager, hand over the assigned activities to the relevant Project Team Members where these differ from the Bidding Team, to ensure continuity into Project Launch and Planning.

The Bidding Team includes roles that are only found within the Bidding stage (such as Capture Leader), roles that are found within both the Bidding and Project execution stages with different accountabilities (such as Project Engineer) and roles that are specific to individual Lines of Business (such as SRS Supply Chain and Procurement Manager).

4.2.2 Bidding Team – Roles specific to the Bidding Phase

4.2.2.1 Capture Leader

The **Capture Leader** is the strategic and commercial leader of the Bids. The Capture Leader role is performed, as required, by the Sales Manager or by other roles designated for the specific bid. In the case of VHC, the role of Capture Leader can be executed by the Bid Manager.

The **Capture Leader** is fully accountable end-to end for the overall success of the Bid - from defining the capture strategy, through securing internal approvals, to submitting the Offer, and then managing negotiations and finalising the contract.

In LoB SRS, the Capture Leader is accountable, with the support of the Bid Manager, for the internal approval process according to the RSBU governance rules.

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In LoB VHC, if the Capture Leader is a separate role from the Bid Manager, the Bid Manager is accountable for managing internal approval with the support of the Capture Leader.

During the bid preparation, the **Capture Leader** is accountable for setting the guidelines for the Bid Preparation, including the definition of Strategic goals (e.g. understanding customer expectations, managing relationships with third parties, identifying potential partnerships or key external providers, and addressing any specific local regulations, liaising with the Contract Manager as needed), target price and key winning factors.

The **Capture Leader** manages the interface with the Customer, directly involving team members whenever necessary. They act as the primary interface towards the Customer, with the support of the relevant Market organisation as well as of the LoB Sales organisation and other functions as applicable.

The Capture Leader remains accountable for the overall Bid, whilst the Bid Manager is accountable for the preparation and consolidation of the Bid according to the RSBU governance rules.

Following the approval and authorization stages, and based on the mandate received, the **Capture Leader** is responsible for leading the negotiations and for finalising the best possible contract for RSBU.

4.2.2.2 Bid Manager

The **Bid Manager (BM)** is accountable for the end-to-end preparation of the technical and economic proposal (including Price, Gross Margin and associated financial parameters) and schedule, ensuring the integrity and consistency of the proposed cross-functional solution. As the inter-functional leader, the Bid Manager coordinates the bidding team (composed of the representatives of all contributing functions) and steers the entire process through to its closure.

The **Bid Manager**:

- ensures that each contributing function (including Engineering, RAM & Safety/RAMS, Quality, Manufacturing, Procurement, Supply Chain, Contract Management, Finance, Legal, Risk Management, Compliance, Sustainability and other relevant Support Functions) formally nominates a representative within the Bidding Team, responsible for taking ownership and accountability of the relevant portion of the Scope of Work;
- provides all the Team Members with the complete customer solicitation documents (Tender, RFP, RFI, Request for Offer, etc) including the bidding SoW and expected outputs (e.g. Deliverables list), and the complete set of customer requirements ensuring that all requirements are properly captured and integrated into the bid execution framework;

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- provides to the bidding team all the relevant information necessary for the definition of the procurement strategy and the Vendor List.
- coordinates all Team Members to ensure structural coherence and integration of the proposed complete solution.
- leads the consolidation of cost estimations and ensures validation of the project whole life costs evaluation, financial assumptions and contingencies, in coordination with the Bid Controller and Project Manager;
- For **Key Initiatives**, Vehicles Bids, and other complex bids, where early execution-driven structuring is required due to execution complexity and long-term delivery exposure, Project Manager leads the identification, assessment, and consolidation of risks and opportunities supported by the Bid Manager. The Bid Manager is accountable for coordination and formal integration of the resulting risk and opportunity profile into the Bid baseline, ensuring full traceability, consistency and alignment with governance requirements.
- For all bids other than **Key Initiatives**, Vehicles Bids, and other complex bids, where complexity or long-term delivery exposure are limited (including standard bids), the Bid Manager leads the identification, assessment and consolidation of risks and opportunities. In these cases, the Bid Manager ensures that all relevant MWPMs, WPMs and WPLs formally identify and assess risks within their assigned scope and consolidates them at Bid level. Bid Manager formally consolidates and integrates the resulting risk profile into the Bid baseline, ensuring full traceability, consistency and alignment with governance requirements.
- collects all the information and contributions from each Bidding Team members in order to consolidate all required documentation to ensure compliance with, and approval under, RSBU Governance.
- consolidates all bid documentation and ensures stage readiness and compliance with authorization rules and internal governance processes.
- secures the integrity and completeness of the bid baseline - technical, financial and scheduling, including whole life costs estimates, assumptions, revenues, margins, Net Present Value (NPV) risks, contingencies and work schedule.
- prepares a structured and traceable handover package to ensure seamless transition to the Project Manager enabling the final review and proper handover from bidding to the project execution stage.

In case of award, the **Bid Manager** is accountable for the complete handover to the Project Manager, and project team (as required) of all bid relevant data (contractual data and baseline elements, assumption and technical requirements, economics evaluations, ...) safeguarding continuity, integrity and accountability from bid to project.

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4.2.2.3 Qualification Specialist

The **Qualification Specialist**:

- ensures the definition of the list and the collection of qualification documents to be prepared for the submission of assigned prequalification/bids;
- ensures the submission of the prequalification package (or qualification documentation), in collaboration with the Capture Leader and, when needed, the Bid Manager;
- ensures the implementation of improvement actions and the maintenance of the database of references and other relevant qualification documentation;
- supports the Capture Leader and the Bid Manager in the responding to clarification requests to/ and from the Customer, related to qualification matters for the assigned bid.

4.2.3 Bidding Team – Roles specific to the Bidding Phase [SRS]

4.2.3.1 Bid Procurement Manager

Tailoring or Adjustments:

These tailoring principles only apply in the case of SRS projects.

Depending on the complexity of the initiative, the Bid Manager may appoint a **Bid Procurement Manager** to support procurement-related activities within the Bid Team.

When appointed, the **Bid Procurement Manager** leads and coordinates bid-phase procurement activities, working with Category Manager and Buyers who manage the sourcing and contracting of individual work packages.

All bid-phase procurement activities are performed in close collaboration with Supply Chain and Procurement Managers [SRS] (SCPM), ensuring alignment with execution strategy, cost structure, and supply chain feasibility. Project execution phase responsibilities remain with the Supply Chain and Procurement organisation.

Role Purpose:

Leads and is accountable for procurement activities during the bidding phase, the Bid Procurement Manager coordinates bid-phase procurement activities with Category Managers and Buyers, ensuring that procurement strategy, external costs, supplier engagement, and risks are aligned with the Scope of Work, technical solution, schedule, and contractual requirements, enabling a competitive and executable bid.

Main Responsibilities:

- leads and coordinate all procurement activities during the bid phase
- ensures Category Managers define the procurement strategy, aligned with bid objectives, procurement rules, and execution feasibility

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- ensures external cost estimates are accurate, complete, and aligned with Scope of Work and contractual requirements
- provides procurement inputs to the bid, including costs, schedules, risks, and assumptions, ensuring that are properly documented
- ensures a structured evaluation of procurement-related risks
- ensures traceability and a compliant handover to the execution phase

Key Tasks

- coordinates Category Managers to define the procurement strategy and preliminary vendor list
- ensures consistency between procurement costs and bid assumptions
- ensures Category Managers validate supplier offers, assumptions, and commercial conditions, including RFQs, clarifications, and bid-phase negotiations
- involves Supply Chain and Procurement Manager in identifying long-lead items, assessing procurement schedule impacts, and estimating additional procurement-related costs (e.g. logistics, spares, duties, escalation)
- coordinates the collection of procurement risks for inclusion in the bid risk register
- ensures Category Managers perform contractual flow-down to suppliers
- coordinates the documentation of procurement scope, assumptions, and commitments, and provide the Bid Manager with a comprehensive Procurement Scope of Work including detailed cost information.
- shares timely the selected procurement strategy with SCPM to allow proper internal-cost estimate for the execution phase.

4.2.4 Bidding Team – Roles applicable in VHC and SRS

4.2.4.1 Project Manager

Tailoring or Adjustments: Only one **Project Manager (PM)** is foreseen within the LoB (work split between LE or Segments will use a WPL structure). For O&M bids this role is covered by O&M Program Manager, for S&M standalone bids this role will be referred to as S&M Program Manager

Main Responsibilities: To guarantee continuity between Bidding and Execution, where possible, the execution **PM** should be nominated during the bid phase, unless the bid does not have the appropriate scale or relevance (as defined by the Bid Manager) to justify the appointment, working on the bid under the leadership of the Bid Manager. Responsibility for the bid remains with the Bid Manager, but the **PM** supports analysis and preparation to smooth the transition into project execution. The **PM** supports the Bid Manager to ensure the proposed solution is executable, planned, costed and risk-assessed for successful delivery and prepared for a structured handover.

Key Tasks:

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- Provides the Bid Manager with hour and cost estimation, as well as associated assumptions and deliverables for all relevant Project Management and Project Management support activities
- Analyses tender specifications (particularly the contractual requirement) from an execution perspective, liaising with CM as needed
- Reviews and captures the non-technical bid requirements to ensure they are incorporated into the Project Manager's deliverables
- Ensures assessment and review of costs validated by the bid manager based on previous project execution
- Validates the Master Plan prepared by the Project Planner
- Participates in the definition of key contractual assumptions
- For **Key Initiatives**, Vehicles Bids, and other complex bids, (where early execution-driven structuring is required due to execution complexity and long-term delivery exposure) the **Project Manager** leads the identification, assessment and consolidation of risks and opportunities during the Bid phase with the support of the Bid Manager. In this context, the **Project Manager** ensures that all relevant MWPMs, WPMs and WPLs formally identify and assess risks within their assigned scope and consolidates them at Bid level, ensuring alignment between risk assessment and execution strategy. The **PM** is accountable for the overall risk and opportunity profile and register, ensuring its robustness, completeness, and consistency with the proposed delivery strategy before finalizing and sharing with the Bid Manager for incorporation into the Bid Baseline. The resulting risk profile is then provided to the Bid Manager who is accountable for formal consolidation and integration into the Bid baseline in accordance with governance requirements.
- For all other bids, where complexity or long-term delivery exposure are limited (including standard bids), the Bid Manager leads directly the risk identification, assessment and consolidation process during the Bid phase.
- Ensures that a delivery strategy is defined in the bidding phase, supporting an efficient mobilisation and contract award handover

SRS

- If BIM (Building Information Monitoring) is required by the bid, ensures that the Employer's Information Requirements (EIR) and all BIM-related tender requirements are analysed; initiates and oversees the preparation of the preliminary BEP

4.2.4.2 Bid Controller

Tailoring or Adjustments:

In the LoB Vehicles, the Bid Controller is known as the Project Controller even within the Bidding phase.

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Main Responsibilities: The Bid Controller (BC) reviews the consolidated bid budget and assists the Bid Manager on economic and financial evaluations. Based on RSBU governance, a given profitability target is identified for each bid and BC responsibilities focus on ensuring this is respected.

Key Tasks:

- Provides the Bid Manager (BM) with its own internal controlling budget for the bid initiative (not cost but quantities/hours), including assumptions and limitations.
- Checks cost estimations with the support of the BM, ensuring consistency with project scope and other comparable projects/bids.
- Performs economic and financial analysis, supporting the BM in cash flow and NPV (Net Present Value) calculations.
- Provides the BM with the financial parameters applied to each Legal Entity for the consolidation of the P&L at LoB level, according to their availability from FP&A.
- Provides the hourly rates used for the cost estimations, based on the official rates existing in the ERP system, and requests to the CFO and COO for approval of any adjustment where needed.
- Checks risk analysis (both contingency and penalties) and contributes to the final review of associated costs and risks associated with the project/bid.
- Supports the BM in the handover process to project execution, formally transferring ownership and accountability of an established baseline, sharing all relevant financial documentation and information to the Project Manager (PM), the Bid Controller and the project team as required.

4.2.4.3 Project Planner

Main Responsibilities: The **Project Planner (PJP)** is accountable for the preparation of the Project Delivery Master Schedule based on contractual requirements and data provided by the Team members/Workload optimisation function, within the timeframe defined by the Bid Manager.

Key Tasks:

- Creates and delivers the Project Delivery Master Schedule to the Bid Manager for final approval after validation by the PM
- In case of Hitachi Rail's inclusion in consortium, the Project Planner is responsible for setting the interface schedule with partners or other stakeholders involved in the project in order to ensure effective delivery of Hitachi Rail's scope of work
- Ensures proper planning documentation required during the bid
- Provides the Bid Manager with its internal Planning Scope of work description (incl. Assumption and limitation) and with its related hours and cost estimation
- Provides schedule narrative, critical path, and other insights/analysis as appropriate

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4.2.4.4 Project Engineer

Tailoring or Adjustments: One Project Engineer within the Bid (properly supported for complex projects by a dedicated team). In the case of multi-entity Bids, where there may be several engineering WPM/WPLs responsible for their respective perimeter of solution, the **Project Engineer (PE)** (MWPM) must be identified who holds the final technical solution authority and is accountable for the entire Bid.

Main Responsibilities: The PE leads and is accountable for the definition of the technical solution and for the preparation, validation and submission of the overall technical proposal in compliance with customer requirements, relevant norms/standards/regulatory frameworks and the assigned project constraints (time, cost, quality)

Key Tasks:

Technical Solution Ownership

- Leads, coordinates, and oversees bid technical solution preparation.
- Coordinates Bid technical team across all engineering disciplines and relevant interfaces and guarantees full integration of each contribution in the overall Technical Solution (which is then further integrated into the overall proposed solution by the Bid Manager).
- Ensures that the bid technical solution and the anticipated development activities are aligned with product roadmap, as defined by portfolio/product line manager.
- Ensures that the production and maintenance/service aspects of the Solution are taken into account in terms of: Manufacturability, Maintainability, Consistency between Non-Recurring and Recurring/Production costs.
 - **VHC:** Accountable for the preparation and maintenance throughout the Bid phase of the Product Data Sheet (PDS), to allow the Quotation Manager to start collecting and consolidating the industrial product cost to be provided to the Bid Manager
- Checks that the Engineering Environment (process, practices and tools) for the Solution is consistent with the project requirements and the applicable engineering environment rules.
- **SRS:** Ensures coordination with Technical Authority for relevant bids and submission of required documents for assessment

Customer and Stakeholder Interface

- Leads technical issue discussions both with the customer and partners, in coordination with Bid team.

Requirement and Compliance Management

- Accountable for ensuring that the technical bid proposal is prepared in compliance with customer requirements, relevant norms, standards, regulatory

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frameworks and internal rules (e.g. Export Control, Environmental/ HSE, Cybersecurity, Safety, RAM).

- Ensures the bid technical proposal is feasible within the assigned constraints (including time, cost, quality, skilled resources availability and industrial organisation)
- Ensures requirements management, including clause by clause assessment of customer specification and preparation of the compliance matrix.
- Drives the preparation of technical specifications to support requests for the suppliers' quotation

Technical Configuration and Baseline Management

- Ensures technical baselines capture and initial PBS preparation.

Technical Risk and Opportunity

- Responsible for providing the Bid Manager with the technical Risks and Opportunities.
- Supports mitigation strategies definition.

Documentation and Deliverables

- Ensures preparation and delivery of required technical documentation.
- Provides the Bid Manager with all relevant hours and cost estimations, including assumptions and limitations of the project, relevant to the proposed technical solution.
- Responsible for the outlining the certification and the validation strategies.
- Responsible for timely submission of the technical proposals to the customer.

4.2.4.5 Contract Manager

Tailoring or Adjustments: In small and low risk projects, the role of the Contract Manager may be held by another member of the project team at the discretion of the BM and functional management

Main Responsibilities: The Contract Manager supports the assessment of the contract, identifying any departures and determining how best to meet the requirements in terms of commercial structure.

Key Tasks:

- Ensures all required due diligence reviews and export and trade control checks have been initiated and conducted on all Project Parties before bid submission and documented accordingly
- Reviews the prime/head contract terms and conditions, supporting the bid manager to formulate a list of commercial and legal departures for bid submission
- Supports the Capture Leader in consultation other LCC team members to formulate and negotiate the terms sheet and/or pre-bid teaming arrangement, along with exclusivity agreements or others

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- Formulate and prepare the flow down terms or principles for major “buy” packages to support Procurement in price sourcing
- Ensure all proposed key subcontractors for major “buy” packages have undergone due diligence reviews
- Supports the Project Team, in particular Project Engineer, PM, and Bid Manager in ensuring the technical and commercial assumptions are appropriately worded and sufficiently robust for bid submission
- The Contract Manager provides recommendations to the Bid Manager regarding the commercial and contractual aspects of the tender
- Provide the Bid Manager with its Contract Management Scope of Work description (including assumptions and limitations) and with its related hours and cost estimation.

4.2.4.6 Project Quality Manager

Main Responsibilities: The **Project Quality Manager (PQM)** is responsible for analysing the tender quality requirements, inputting into **PQM** work/cost estimations, and ensuring quality requirements are flowed down into supplier contracts

Key Tasks:

- Ensures the analysis of the tender quality requirements
- Ensures the estimation of the planned quality assurance activities scope of work (including assumptions and limitations) and the hours/commitment expected during the project execution phase (to implement the QMP and associated manuals)
- Monitors, in coordination with procurement, that identified suppliers are aligned with the quality requirements (across the full scope of work) and that they are on the supplier register
- Ensure traceability and homogeneity of bid deliverables (decision, risks, documents, etc.)
- Assesses bid quality performance and maturity by delivering Quality Advice (Q-Advice) and raising alerts as applicable
- Ensure lessons learned are used at the start and collected at the end of bid

4.2.4.7 HSE Referent

Main Responsibilities: The **HSE tender referent** is responsible for the Preliminary HSE Plan issue (when required)

Key Tasks:

- Ensures that all specific HSE requirements (regulatory and contractual) are identified, shared among Bidding Team and addressed in the tender deliveries
- Ensures support to the Bidding Team to comply with HSE regulatory and project requirements

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- Ensure identification of HSE requirements for suppliers and contractors and deliver HSE Purchase Technical specifications
- Identify the HSE Bid deliverables (HSE Preliminary Plan, HSE KPIS, HSE Management System, etc.) and ensure its delivery
- Prepare the most cost-effective HSE estimation including internal resource hours, travelling requirements, appropriate allocation of local or consultant HSE resources in line with legal and contractual requirements
- In the case of a Consortium or Joint Venture with other entities (e.g., Civil Contractors or Hitachi Group parent companies), support the correct assignment of HSE responsibilities within the Scope of Work (SOW) agreement and highlight any potential weaknesses, risks, or missing role assignment.

4.2.5 Bidding Team – Roles applicable in SRS

4.2.5.1 Project RAMS Manager

Tailoring or Adjustments: The **Project RAMS Manager**, while reporting quality, time, and cost to the BM, acts independently and without influence to ensure safety decision-making.

Main Responsibilities: The **Project RAMS Manager** ensures all RAM and Safety requirements are clearly identified, assessed for their applicability, and allocated across each subsystem and main assembly levels to ensure a safe and compliant operation

Key Tasks

- ensure the analysis of eventual tender RAM, FRACAS, LCC, Safety and Human Factor requirements to identify any relevant laws, standards, company processes, regulations, and certification requirements
- ensure the definition of RAMS targets, strategy and analysis for the integrated system/vehicle and any relevant sub-system or assembly
- ensure the definition and alignment with the Independent Validation strategy
- ensure the coordination and confirmation of the RAMS SoW across different entities (in line with safety maturity level roadmap) for bids involving RAMS WPLs from multiple entities
- coordinate with the relevant engineering referent to receive information on contractual technical gap analysis and preliminary proposed design solution as well as support the identification of improvement opportunities for system design/vehicle configuration for RAMS scope
- support the transfer of RAMS requirements list to all potential suppliers as well as technically evaluate the offers from potential suppliers for RAMS-related performance

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- ensure the preparation of relevant RAMS tender documentation and coordinate with external partners for establishing the appropriate safety strategy (according to tender requirements)
- provide the Bid Manager with RAMS and Independent Validation Scope of Work description (including assumptions and limitations) and related estimated hours and cost (including training, assessment bodies, travels etc.).

4.2.5.2 Supply Chain and Procurement Manager

Tailoring or Adjustments: n/a

Main Responsibilities: Supports the Bid Manager in assessing supply chain and procurement requirements for the bid, ensuring accurate cost and resource estimates based on initial assumptions

Key Tasks:

- participates in the bidding procurement strategy definition, in compliance with company's procurement guidelines
- evaluates the make or buy strategy, verifying that all external cost estimation matches with project requirements and contractual requirements
- supports material estimation for major assemblies and equipment
- records make vs buy/procurement strategy decisions in the make and buy cost estimate sheet
- identifies any Long Lead items that the project may need to procure prior to Notice to Proceed or as soon as the contract is signed, to ensure that project milestones are met
- contributes to the development of the project programme, advising on lead times for material procurement
- contributes to the risk register, advising on relevant risks associated with procurement activities
- evaluates the bid, capturing any key contract requirements regarding procurement, particularly those relevant to tender evaluation criteria
- supports the project engineer in analysis of obsolescence information on bid technical solution
- provides the Bid Manager with its Supply Chain & Procurement Scope of Work description (including assumptions and limitations) and with its related hours and cost estimation.

4.2.5.3 Construction Manager

Main Responsibilities: The **Construction Manager** is responsible for reviewing project requirements and defining the construction and installation strategy to support the bidding process. The role ensures that all construction-related inputs (including scope, costs, and risks) are complete, accurate, and aligned with project requirements. The **Construction Manager** defines the construction portion of the

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project schedule, including milestones and dependencies, and ensures overall feasibility from both constructability and execution perspectives.

Key Tasks

- reviews the project requirements related to construction/installation activities, and identifies constructability constraints or missing inputs impacting schedule or budget
- performs site surveys and prepares a formal Site Survey Report, including risks and opportunities related to physical site conditions and predecessor activities
- defines the construction schedule, including key milestones, mobilization dates, installation durations, and interfaces, and shares it with the Project Planner who validates logic and format before they integrate it into the master schedule
- defines construction execution assumptions and exclusions to ensure transparency in the bid.
- delivers the Procurement Technical Specification relevant to construction activities to the SCPM in order to define the scope of work along with the guidelines, technical provisions, and requirements that the external provider will have to respect
- defines the construction/installation strategy, team organisation, and workload (internal and outsourced), as well as the site mobilization plan for construction/installation activities.
- provides the Bid Manager with its Construction Scope of Work description (including assumptions and limitations) along with the related hours and cost estimation
- provides technical support to the Procurement Department in the identification and selection of installation subcontractors and supports in the technical review and evaluation of the quotations. Reviews subcontractor bids for technical completeness and alignment with the construction strategy.

4.2.5.4 Commissioning Manager [SRS]

Main Responsibilities: Responsible for preparing the commissioning strategy and commissioning schedule in alignment with the overall project master schedule.

Key Tasks:

- Ensures the definition of the commissioning strategy and of the commissioning activities, as well as the delivery of the Commissioning Plan
- Provides the Commissioning Work Schedule to the Project Planner to integrate the overall schedule with the detailed commissioning activities, according to the Commissioning plan and Verification and Validation (V&V) Plan.
- Ensures the costs and resource estimation regarding the on site testing activities, including assumptions and limitations as well as commissioning coordination efforts

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4.2.5.5 Warranty Manager [SRS]

Main Responsibilities: Creates end-to-end service model, warranty strategy and concept; ensures warranty concept and commitments are compliant to customer requirements. Provides resource and material estimates associated with the completion of warranty obligations and ensures preparation of technical deliverables or any needed documentation

Key Tasks

- ensures the maintenance/warranty requirements analysis and allocation for the bid as per tender documents
- ensures the preparation of any warranty bid technical deliverables in line with bid schedule and tender requirements
- ensures the alignment of the warranty scope with technical baseline, service strategy and supplier back-to-back coverage
- ensures the analysis and approval of supplier contribute to warranty and service (coverage, duration, advance replacements)
- ensures the alignment with warranty policy and legal/regulatory requirements.
- estimates (with the support of RAM data) any material necessary for the execution of the warranty period and design the correct approach to the KPIs analysis and demonstration
- provides the Bid Manager with its Warranty management Scope of work description (incl. Assumption and limitation) and with its related hours and cost estimation.

4.2.6 Bidding Team – Roles applicable in VHC

4.2.6.1 Project RAM Manager

Main Responsibilities: The **Project RAM Manager** is responsible for the identification of the RAM requirements of the tender and assesses their applicability, as well as allocation to each subsystem/main assembly.

Key Tasks:

- ensures the analysis of required RAM and LCC performance
- defines RAM and LCC targets, strategy and analysis for the integrated system/vehicle and any relevant sub-system/assembly
- co-ordinates with the PE to receive information on contractual technical gap analysis and preliminary proposed design solution
- supports the PE in identification of improvement opportunities for system design/vehicle configuration for RAM and LCC scope
- supports transfer of RAM and LCC requirements list to all potential sub-suppliers
- ensures the preparation of relevant RAM and LCC tender documentation

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- identifies material requirements to support reliability
- supports the Risk manager in the identification of potential risks and mitigation measures
- provides the Bid Manager with its RAM Management Scope of Work description (including assumptions and limitations) and with its related hours and cost estimation
- ensures identification and escalation of design config critical aspects

4.2.6.2 Project Safety Manager

Tailoring or Adjustments: Although reporting to the BM, the **Project Safety Manager** operates in an independent way in accordance with CENELEC/IEC standards

Main Responsibilities: The **Project Safety Manager** is responsible for the identification of the Safety requirements of the tender and assesses their applicability, as well as allocation to each subsystem/main assembly

Key Tasks:

- ensures the analysis of the tender safety requirements to identify any laws/regulations for safety approval, and estimating the relevant costs
- defines safety targets, strategy and analysis for the integrated system/vehicle and any relevant sub-system/assembly
- co-ordinates with the PE to receive information on contractual technical gap analysis and preliminary proposed design solution
- supports the PE for identification of improvement opportunities for system design/vehicle configuration for Safety scope
- supports transfer of Safety requirements list to all potential sub-suppliers
- ensures the preparation of relevant Safety tender documentation
- provides the Bid Manager with its Safety Management Scope of Work description (including assumptions and limitations) and with its related hours and cost estimation
- ensures the identification of any missing safety requirements
- ensures identification and escalation of design configuration critical aspects.

4.2.6.3 Project Procurement Manager

Main Responsibilities: The **Project Procurement Manager (PPM)** leads and is accountable for procurement activities during the bid phase, ensuring contractual procurement requirements are understood, sourcing strategy is defined, supplier assumptions are credible, and external costs are robust to support a compliant and deliverable offer

Key Tasks:

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- ensures analysis of contractual requirements relevant for procurement, in alignment with specifications from stakeholders and with the support of Legal and Project Management, prepares the RFP/RFQ
- ensures the definition of the vendor list in collaboration with stakeholders
- ensures quoted materials are in accordance with contractual milestones and technical, cost, and performance requirement
- coordinates the management of suppliers, coordinating supplier interactions during the bid and ensuring supplier offers meet technical, performance, schedule and commercial requirements
- ensures the definition and update of the bid procurement plan
- contributes to the supplier assessment process
- ensures the definition of a list of major cost items, highlighting any deviation from the baseline value, supporting the Design-To-Cost activities where appropriate
- contributes to risk analysis for procurement and supplier risks (including lead time, capacity, single source exposure, commercial constraints, etc) and proposing mitigations and contingencies
- supports the quotation manager in definition of make/buy mix by defining the costs for all the proposed configuration
- provides the Bid Manager with its Procurement Scope of Work description (including assumptions and limitations) and with its related hours and cost estimation.

4.2.6.4 Test Manager

Main Responsibilities: Ensures the analysis and feasibility of any testing requirements (tender and internal).

Key Tasks:

- supported by the PE, develops an efficient plan for the type testing of vehicles, which will ensure the development, testing and validation of the project (according to the Project Master Schedule)
- manages the overall type testing plan incorporating components (under the responsibility of the Project Engineer) and Vehicles
- ensures the analysis and the management of the involvement of any eventual external infrastructures required by the experimental vehicle tests
- ensures the analysis of the test tender requirements – technical and delivery methods/constraints
- ensures the analysis of the standard applicable to the bid received by the Certification and Safety Manager
- ensures that, with close cooperation with the Project Engineer and relevant WPLs, an effective and efficient plan is developed for the Type test, Routine Testing and Acceptance test of the project

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- contributes to the risk matrix about testing for the project analysed
- is responsible for developing an estimate of cost that takes into account all that is required to support for the Type test, Routine Testing and Acceptance test of the project or any other support required by the project. This includes all external/third party costs for infrastructure required by experimental vehicle testing.
- maintains the relationship with internal and external stakeholders (notified body laboratory and National Safety Authority) necessary for proper cost estimation
- highlights any required changes or any need of investments in the vehicle or component test facilities (including simulators)

4.2.6.5 Project Operations Manager

Tailoring or Adjustments: Project Operations Manager (POM) for Rolling Stock (RS) project are the former Workload Optimisation and Balancing (WOB) and Project Manufacturing Manager referents. **POM** for Service and Maintenance (S&M) project is the former S&M Supply Chain Manager

Main Responsibilities:

RS: define and foresees the relevant workload balancing and industrial activities that will impact the plants

S&M: defining the spare parts stock and leading repairs

Key Tasks:

Rolling Stock (role is usually split across more than one referent: a WOB specialist and a Plants Industrialisation specialist)

- supports the definition of the most competitive solutions (technical solution, make or buy decision, etc), providing all relevant elements related to manufacturing, feeding in the required manufacturing milestones that will be included in an Integrated Plan as an output measure for other Functions
- defines and agrees the manufacturing scope of work among the production sites and production plan (to be shared with Project Planner)
- ensures the definition of the manufacturing processes and cycle times to be put in place for the production of the vehicles
- confirms with the other stakeholders (primarily PE and PPM) the manufacturing feasibility assumptions
- estimates the costs, and durations needed for manufacturing jigs, learning curve and production hours, and transmits them to the Quotation Manager, declaring assumptions and risks.

Service and Maintenance

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- Defines the labour model (workforce size, workload, skills) based on the maintenance plan
- Defines spare parts strategy, including initial provisioning and consumption curves
- Estimates costs, durations, learning curves, risks and opportunities, providing structured inputs to the Quotation Manager
- Ensures cross-functional alignment with Engineering, Operations, Industrialisation and Supply Chain during the bid phase

4.2.6.6 Commissioning Manager [VHC]

Main Responsibilities: Responsible for preparing the commissioning strategy and commissioning schedule in alignment with the overall project master schedule.

Key Tasks:

- Provides the Commissioning Work Schedule to the Project Planner to integrate the overall schedule with the detailed commissioning activities, according to the V&V plan
- Responsible for defining the commissioning strategy and assessing/validating that the costs and resource estimation done by the WPLs regarding on site testing activities are compliant with the strategy
- Provide the Bid Manager with its Commissioning Scope of work description (incl. Assumption and limitation), with the estimation of the commissioning coordination costs and validating commissioning activities costs
- Identifies risks and opportunities within commissioning activities

4.2.6.7 Warranty Manager [VHC]

Main Responsibilities: Provide resource and material estimates associated with the completion of warranty obligations and support preparation of technical deliverables or documentation

Key Tasks:

- ensures the maintenance requirements analysis and allocation for the bid as per tender documents
- ensures the preparation of any warranty bid technical deliverables in line with bid schedule and tender requirements
- with the support of RAM data, estimates material necessary for the execution of the warranty period
- provides the Bid Manager with its Warrant Management Scope of work description (incl. Assumption and limitation) and with its related hours and cost estimation
- identifies risks and opportunities within warranty activities.

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4.3 Project Execution Team

4.3.1 Project Execution Team – Standard Principles

The Project Execution Team (hereafter referred to as Project Team) is the inter-functional organisational structure responsible for managing the Project under the leadership of the Project Manager, during the whole project lifecycle. A Project Team is set up when a new contract is awarded to RSBU and a project has to be launched in order to execute the relevant scope of work.

The composition of the Project Team shall be defined and aligned with the structure outlined in this document. In case of specific customer requests, a different additional team representation can be created specifically for external use only, with no impact on the way the RSBU manages and delivers the project.

Although the Project Team is expected to be staffed in accordance with this document, due to the variety in size and complexity of project execution, the following principles may be taken in consideration for tailoring:

- A Project Team role does not necessarily correspond to 1 Full Time Equivalent (FTE). Based on workload and complexity, the Project Manager and the relevant functional organisation determine the appropriate FTE allocation
- In certain cases (e.g. small and less complex projects with highly experienced professionals), one individual may cover more than one role (either fully or partially), subject to joint assessment by the Project Manager and the relevant functional organisation. In such cases, the individual ensures appropriate functional reporting within the functional organisations

Project Team Members are responsible for performing the assigned activities on budget, quality and on time, and in line with Safety >> Quality > Delivery > Cost (S>>Q>D>C) approach, according to the relevant plans/schedule and to the project baseline.

The Project Team Members are responsible for the management and control of the assigned project activities within RSBU and are delegated full authority by the organisation to do this within the project context.

In addition to its role-specific responsibility, each Project Team Member is responsible to:

- issue all the documentation relevant to their functional area
- provide the Project Planner and Project Manager with the related detailed project delivery schedule and periodical physical progress
- provide to the Project Controller (via any applicable tool or system) a periodical progress of the hours, costs and requirements related to the execution of the activities within the assigned SoW / WPs
- provide the Project Manager with the evaluation of the risks and opportunities related to the assigned SoW

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The Project Team operates transversally to RSBU functions, business segments, regions and Legal Entities and collaborates towards a joint project outcome, either successful or unsuccessful. As such, the Project Manager is accountable for the performance of the assigned activities (and control of associated costs and margin) at project level, regardless of the Legal Entity/ies where they are executed and generated. In this respect, intra-company orders, necessary to formalise the relationship among RSBU Legal Entities, have an administrative purpose.

In order to ensure effective delivery, it is critical to ensure that a single individual maintains ultimate accountability even in complex cross Business Segment, or cross Legal Entity projects. As key principles:

- A single Project Manager is appointed for each project and leads the Project Team and all activities across functions, business segments, regions, and legal entities. The Project Manager holds full end-to-end accountability for the Project, in line with S>>Q>D>C approach.
- A single Project Engineer (PE) acts as the sole technical lead on the project solution. The PE ensures end-to-end technical coherence, integrity and feasibility of the proposed solution.
- In the event that additional project management or engineering roles are required due to the distribution of project activities, these roles shall take the form of WPMs or WPLs and be strictly subordinate to the Project Manager/Project Engineer.

4.3.2 Project Execution Team – Roles applicable to SRS and VHC

4.3.2.1 Project Manager

Tailoring or Adjustments: Only one **Project Manager (PM)** is foreseen within the LoB (work split between LE or Business Segments will use a work package structure)

Main Responsibilities: The **PM** has the overall cross-functional accountability for the end-to-end delivery of the project from contract award to completion, meeting contractual requirements and delivering on safety, budget, quality and time, ensuring customer satisfaction. The **PM** is accountable for risk management and project contingency allocation as well as the customer relationship. The **PM** is ultimately accountable for overall project Profit and Loss (P&L).

Key Tasks:

- drives the definition/review and approval of project baseline, ensuring update and alignment with all the involved stakeholders (internal and external)
- validates and manages any changes to the project baseline
- ensures compliance with contractual requirements (including cost, time and quality) in collaboration with Project Team members
- leads mobilisation and execution of the delivery strategy
- manages risk, opportunity, and contingency for the whole project

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- validates economical and financial progress reports including Revenue, Gross Margin, Invoicing and Cash-in
- prepares and monitors, with Project Controller, the annual project budget and quarterly updates (forecasts)
- manages customer communications and relationships, including variation requests, supported by Contract Management as needed
- validates sub-contracts/letters of intent relevant to key sub-suppliers and RSBU Legal Entities (including cost, quality and schedule)
- ensures final authorisation of any deliveries to the customer
- monitors and manages all work packages through prescribed project management governance MPR (Monthly Project Review), escalating through QPR (Quarterly Project Review) as required, ensuring project team members staffing is correct
- where appropriate, ensures the proper project transition from Design and Build (DB) to an Operation and Maintenance (O&M) or Service & Maintenance (S&M) Project Manager
- manages and leads the project team
- manages external and internal stakeholders
- ensures lessons learned and drives projects execution towards Continuous Improvement (CI)

SRS

- If Building Information Modelling (BIM) is required by the project, being the overall BIM Process Owner, the PM initiates and oversees the preparation of the BIM Execution Plan and ensures its full implementation

4.3.2.2 Project Controller

Main Responsibilities: The Project Controller is responsible for guiding and driving the project and for the control of its economic and financial aspects. In particular, this role is accountable for:

- monitoring the financial aspects of the project, including order intake, revenues, estimate at completion (EAC), backlog, backlog profitability and workability, inventory analysis, workflow analysis, and working capital.
- reviewing, questioning and challenging these metrics with the Bid/Project Manager and the relevant team members, ensuring robustness of the financial picture.
- monitoring the project costs and revenues, supporting the PM in the analysis of deviations from the baseline;
- defining the corrective actions and leading their tracking activities;
- reporting and consolidating the financial data at project level;
- verifying and preventing any risk of extra costs impacting the Whole Life Budget (equal EAC);

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- proposing updates to the contingencies;
- analysing and discussing with the business the blocked trade receivables;
- tracking and highlighting profitability improvement opportunities;
- notifying to the FAS&T and CFO any change in the EAC.

Key Tasks:

- Maintains the project baseline for financial aspects and monitor deviations, trends, and cost to complete.
- Monitors and reports economic progress (forecasts, yearly budget and budget-plan) including at project reviews and for internal Company reporting.
- Identifies, tracks and reports profitability improvement opportunities and financial risk to the Project Manager.
- Co-ordinates support/contributions relating to the project from the whole CFO organisation.
- Registers and officialises (upon management approval) any variations to baseline.
- Reports and consolidates the project level data, including the order intake, backlog, utilising a BPC tool.
- Reports and monitors WIP and Working Capital, highlighting risks and required actions.
- Within LoB VHC: the PJC is responsible for developing the WBS (with the support of the Project Manager) and the associated CBS under ultimate accountability of the PM.

Within LoB SRS: the PJC supports the Project Planner who is responsible to develop the WBS, the PJC is responsible for developing the associated CBS under ultimate accountability of the PM.

4.3.2.3 Project Planner

Main Responsibilities: The **Project Planner (PJP)** ensures continuous and active monitoring of the project physical progress and its critical path and monitors the compliance of the project with all contractual milestones. They own project physical progress planning and schedule control during execution, ensuring the baseline is robust, progress is measured credibly, impacts are analysed, and recovery plans are developed

Key Tasks:

- Preparation and Maintenance of Project Master Schedule in line with the contractual WBS
- In case of Hitachi Rail's inclusion in consortium, the Project Planner is responsible for setting and maintaining the interface schedule milestones with partners or other stakeholders involved in project execution in order to ensure effective performance of Hitachi Rail's scope of work

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- Creation and monitoring of the Project Second Level Schedule (based on the contributions from each Project Team member) showing detailed project activities (including engineering, procurement, production, Quality Assurance (QA), safety, certification, delivery, test) including all regions/countries and business segments, along with contractual milestones
- With the support of relevant functions, mitigates any risk of delays impacting contractual obligations, escalating any significant deviations to the Project Manager
- Supports the overall identification, reporting and mitigation/achievement of schedule risks and opportunities
- Ensures regular updating of monitoring, reporting, and analysis of project physical progress with all functions involved aligned to the project plan as a single point of truth

SRS

- The Project Planner is responsible for developing the WBS with the support of the Project Controller and under the ultimate accountability of the PM

4.3.2.4 Project Engineer

Tailoring or Adjustments: One Project Engineer within the Project (properly supported for complex projects by a dedicated team). In the case of multi-entity Projects, where there may be several engineering WPMs/WPLs responsible for their respective perimeter of solution, the **Project Engineer (PE)** (MWPM) must be identified who holds the final technical solution authority and is accountable for the entire Project.

It is important and strongly recommended, whenever feasible, for the same Engineering resource to be appointed in the Bidding and Project Execution phases to ensure business continuity, reinforce ownership, and enhance accountability.

Main Responsibilities: The **Project Engineer** leads and is accountable for the overall technical solution (which covers all contractual technical specification requirements and any additionally generated technical requirement during execution), for its design, delivery and verification and validation, and for the acceptance of the Customer; this by ensuring integration, compliance, and configuration control and according to the budgeted cost, schedule, and quality target of the project. This technical solution is then further integrated into the overall project solution under the accountability of the PM.

Key Tasks:

Technical Solution Ownership

- Leads, coordinates, and oversees the design and delivery of the technical solution.
- Accountable for ensuring that the technical solution is planned, designed, developed, verified and validated, and delivered effectively, efficiently, with

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good quality and in compliance with customer requirements, relevant norms, standards, regulatory frameworks and internal rules (e.g. Export Control, Environmental/ HSE, Cybersecurity, Safety, RAM), within the assigned constraints (time, cost, quality) and in alignment with Product portfolio roadmap and Make-Team-Buy strategy.

- Coordinates project technical team across all engineering disciplines and relevant interfaces and guarantees full integration of each contribution in the overall technical solution development and delivery.
- Ensures that the technical solution and the development activities are aligned with product roadmap, as defined by portfolio/product line manager.
- Ensures that the production and maintenance/service aspects of the Solution are taken into account in terms of: Manufacturability, Maintainability, Consistency between Non-Recurring and Recurring/Production costs.
- Checks that the Engineering Environment (process, practices and tools) for the Solution is consistent with the project requirements and the applicable engineering environment rules.
- **SRS:** Ensure coordination with Technical Authority for relevant projects and submission of required documents for assessment at the start of the project and in case of major change affecting technical solution architecture and reuse strategy

Customer and Stakeholder Interface

- Acts as the primary technical interface with the customer
- Leads technical discussions both with the customer and partners to ensure expectations and questions are addressed appropriately.

Engineering Planning, Governance and Progress Monitoring

- Plans and coordinates engineering activities and deliverables, including development lifecycle selection and implementation strategy, covering entire technical solution.
- Defines the system development cycle and specifies the process, practices and tools applicable to the product / project based on the management system and tailored based on project specific constraints (e.g. tools imposed by the Customer). Those elements are reported in the corresponding Engineering Plans.
- Ensures global consistency of the engineering process, practices and tools and approves lower level solution elements' engineering plans including those of subcontractors and partners.
- Support the functional organisation in identifying the engineering skills needed to ensure Project / Product delivery.
- Responsible for the outlining the certification and the validation strategies.
- Monitors physical progress and performance of all engineering activities, escalating issues to the PM, PQM and functional management, where required.

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- Approves the main engineering reviews and deliverables,
- Organizes appropriate technical meetings and peer reviews, if required, in order to perform the solution technical assessment and validation.
- Ensures the preparation, organisation and reporting of Technical Gate reviews (e.g. Design reviews), internal and external, as well as the management of technical non-conformances as required.
- Reports status and provides input to Project Gate reviews.
- Responsible for timely submission of required technical documentation to the customer.
- Guarantees that the verification of technical activities and documents is performed.
- Ensures the applicable Engineering Performance initiatives are properly deployed in their Project / Product.
- Support internal and external process audits.

Technical Risk and Opportunity

- Identifies, monitors, and escalates technical risks and opportunities.
- Supports mitigation strategy.

Supplier/Partner Technical Coordination

- Ensure technical specifications are properly cascaded to suppliers.
- Support co-engineering activities as needed.
- Drives the preparation and authorizes the purchase technical specifications
- Ensures that technical and process requirement are identified in SoW and flown down to suppliers and partners.

Requirement and Compliance Management

- Leads and ensures requirements definition and management throughout the entire project lifecycle, by defining Requirement management strategy and properly maintaining the Requirements Traceability Matrix (RTM), duly tracing requirements verification and validation, and relevant evidence of compliance.

Verification and Validation and Quality

- Ensure that V&V activities are defined, executed, and traced.
- Guarantee the quality and accuracy of technical documentation and process compliance.

Technical Configuration and Baseline Management

- Ensure assessment of potential gap from Bid submission at the start of the Project.
- Ensure PBS prepared at Bid phase is confirmed or updated at the start of the project and maintained up to date in appropriate PLM.
- Leads Configuration Items selection, based on PBS validated at the start of the project.
- Ensures proper and timely management of project technical configuration,

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- Ensures control of configuration and technical baselines, authorizing changes to baseline and guaranteeing compliance with contract requirements and applicable standards.
- Enforces the alignment of the technical solution with the baseline assumptions.
- Manages the technical change requests, variation order, non-compliances, and leads change control boards (CCB) to ensure change impact analysis on scope, technical solution, schedule, cost, and compliance is performed, and authorizes or rejects changes with appropriate records.

Documentation and Deliverables

- Plans: Ensure preparation of Engineering Management plans and their consistency with relevant dependent plans
- KPIs: tracks performance and competitiveness KPI, performs root cause analysis of problems and conduct associated corrective actions
- Schedule: Coordinate preparation and updates of project schedule for technical activities; monitors key technical dependencies.
- Documentation and records: Ensures preparation and delivery of required technical documentation as well as reviews and decision records, with support from technical team and relevant Work Package leaders.
 - Plans: the project engineering management plan, the relevant design management plans; the design deliverables plan, interface management plan, configuration management plan, requirement management plan, verification and validation management plan, BIM management plan (if applicable)
 - Technical documents: Complete list of technical documents/artifacts must be defined and managed in PLM, including but not limited to Requirements, design, verification, validation documentation, as well as all relevant manuals, documentation, and training to the customer.
 - Technical and configuration baselines and relevant data packs.
 - If BIM is required by the project:
 - implements BIM Manager (the Project Manager) guidelines into operational BIM workflows. Ensures the issue of all the BIM deliverables, including BIM management plan, planning (4D) and cost estimation (5D). Manages the federation of multidisciplinary models ensuring geometric and information consistency across disciplines
 - SRS: Supports preparation of the as-built model and final handover deliverables, coordinating model updates during construction and ensuring alignment with site changes.

4.3.2.5 Contract Manager

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Tailoring or Adjustments: In small and low risk projects, the role of the **Contract Manager** may be held by another member of the project team at the discretion of the PM and functional organisation.

Main Responsibilities: The **Contract Manager** ensures that, from contractual and commercial perspectives, contract requirements, responsibilities, and exposures are managed to optimise potential financial return whilst mitigating commercial risk during project delivery.

Key Tasks:

- masters contractual documentation T&Cs, leading the project team in analysis and interpretation and ensuring coverage of any compliance recommendations relative to T&Cs as the output of due diligence reviews
- supports the complete review of the Project during the planning phase
- leads and supports the identification of customer contractual conditions to be flowed down into sub-supplier contracts, support in the engagement of major suppliers to ensure appropriate risk transfer
- ensures the maintenance of contractual records and documentation
- ensures monitoring and reporting on commercial risks and support identification of opportunities and risk mitigation strategies to preserve key project milestones
- supports the management of major sub-contracts, including ensuring early due diligence on key subcontractors
- supports with contract management activities during project delivery
- supports the PM in managing contractual relations with the customer, partners, JV or consortium
- supports the PM with Variation Offers or claims, assessing contractual impact and supporting negotiation following relevant approval
- analyses customer claims or complaints

4.3.2.6 Project Quality Manager

Main Responsibilities: The **Project Quality Manager (PQM)** is responsible for the definition, implementation, and maintenance of the Project Quality System (Quality Plan and Procedures) to ensure all Contractual Quality requirements are met.

Key Tasks:

- acts as the QA authority within the Project Team, defining, implementing and maintaining the Project Quality Management Plan (PQMP), ensuring alignment between project execution and GBMS processes
- support procurement in supplier selection, monitor, in coordination with procurement, that the identified suppliers are aligned with the quality requirements

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- ensures project quality deliveries are compliant with quality requirements
- assesses project quality performance and maturity by delivering Quality Advice (Q-Advice) and raising alerts as applicable
- in collaboration with the project team, monitors the full completion of all quality assessments / process gates in alignment with safety, quality, schedule and cost
- guarantees that Quality and Safety audits (internal and external) are performed on the project according to RSBU standards
- monitors and follows up on non-conformities and related corrective actions (including customer comments)
- conducts Customer Satisfaction surveys at appropriate project stage as agreed with PM
- monitors and follows up improvement action plans as per Customer Satisfaction survey results
- ensures proper analysis, escalation, and management of important quality issues across processes, products and systems installed or delivered to customers
- acts as the direct interface with the Customer, Partners, and External Providers with reference to Quality Assurance aspects
- identifies and supports the management of Quality risks and opportunities
- ensures the completion of Project Lessons Learnt Monitoring
- acts as one of decision makers at engineering technical Gate reviews (e.g. Design reviews), ensuring maturity of solution at each gate as per gate objectives throughout the development life cycle
- **VHC & OS&M:** At the operational site, ensures management of FAI & monitors completion of product and service nonconformity management, compliance checks and external provider quality control

4.3.2.7 HSE Referent

Main Responsibilities: The **HSE Project** referent ensures the identification, monitoring and management of the Occupational Health and Safety and Environmental risks on site, maintaining a work environment compliant with HSE local legislations, International Standards, Corporate and contractual requirements, preventing injury, illnesses, and pollution. The **HSE referent** is responsible for the HSE Plan issue (when required)

Key Tasks:

HSE Planning and Governance

- prepares and updates the Project HSE Plan ensuring alignment with GBMS, contractual, and legal requirements
- defines and communicates project-specific HSE rules as required
- defines Project HSE rules, procedures, and instructions as required.

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Risk Management and Method Statements

- identifies, assesses, and records HSE risks and opportunities in the HSE Risk Register, ensuring it is periodically updated to reflect project evolution and findings from field observations
- reviews Method Statements to ensure HSE hazards are identified and risk controls are adequate and effective

Operational Support and Compliance

- provides HSE advice and support to Project Team to fulfil their responsibilities on HSE
- verifies compliance of project activities, including sub-contractors, with applicable laws, contractual obligations, and project HSE procedures

Monitoring, Reporting and Performance Evaluation

- performs routine HSE inspections, internal audits, and HSE legal/contractual compliance checks
- defines necessary corrective actions, assign owners, and verify timely closure.
- prepares project HSE reports (internal and client-facing), including KPIs, monthly statistics, trends, and dashboards
- VHC: ensure the management of Vehicles updates in conjunction and alignment with the Project Engineer & Project Operations Manager

Incident Management and Emergency Preparedness

- leads or supports incident reporting and investigation processes, ensuring timely communication and action tracking
- coordinates and monitors emergency preparedness, including drills and related performance reports

Training, Competence and Engagement

- identifies training and competence HSE requirements for all project personnel and coordinates/delivers required HSE training, site inductions, toolbox talks, and awareness campaigns (including wellbeing initiatives).
- communicates safety alerts, lessons learned, and best practices across the project.
- participates in HSE coordination meetings with clients, contractors, and internal stakeholders.

4.3.3 Project Execution Team – Roles applicable to SRS

4.3.3.1 Project RAMS Manager

Tailoring or Adjustments: The **Project RAMS Manager**, while reporting quality, time, and cost to the PM, acts independently and without influence to ensure safety decision-making.

Main Responsibilities: The **Project RAMS Manager** ensures the RAM and Safety requirements of the project are clearly identified and correctly applied across the

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system, component, assembly levels, ensuring the delivered solution meets its RAMS requirements to ensure safe and compliant operation

Key Tasks

- coordinates RAMS and Independent Validation WPLs, including the monitoring and follow up on schedule, remedial actions related with physical and economical progress to be agreed with PM and PJC
- ensures the analysis and definition of RAM, FRACAS, LCC, Safety and Human Factor requirements (where applicable) based on company processes, contract, laws, standards, regulations, or any other sources applicable
- ensures the elaboration, review and update (as well as authorizes in coordination with the WPLs), RAMS and Independent Validation documents including the documentation supporting given approvals or deployment scope (e.g. authorisation for testing or RAMS readiness for Operational handover), as well as the coordination of suppliers' RAM and Safety activities
- supports design WPL in confirming the proposed design solution and managing any design changes
- supports the Warranty Manager in the analysis of data from service
- ensures the definition and application of the independent validation
- ensures the certification (and homologation if applicable) strategy implementation; with the support of the relevant stakeholders (e.g. engineering WPLs), interfacing with relevant authorities, aligning external stakeholders and safety approvals/product acceptance/authorisations with project schedule
- exercises escalation and veto rights where deliverables are insufficient, ensuring project decisions uphold required independent validation, safety and reliability levels
- interfaces with the client and relevant project stakeholders (e.g. assessors), involving relevant RAMS and validation WPLs when needed, regarding the management of RAMS activities

4.3.3.2 Supply Chain and Procurement Manager

Main Responsibilities: The **Supply Chain and Procurement (SCPM)** is responsible for the project material flow according to the Project Master schedule and securing the timely involvement of all necessary subcontractors to support procurement, logistics, and delivery of on-site activities

Key Tasks

- develops the Supply Chain Master Schedule (including PO definition, materials delivery, external costs etc.) matching the Project Master Schedule and the site activities schedule

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- estimates risks and opportunities against the Project Master Schedule both in terms of Procurement (Buy) and Manufacturing (Make)
- launches production and activates procurement on time, synchronizing Supply Chain activities and acting as a unique point of contact for manufacturing, logistics, and purchasing
- monitors and expedites vendors' and factory activities to meet targets on milestone achievement (raw material availability, FAT, delivery on site) and external costs (work progress, actuals, invoicing expectation, forecast)
- activates and coordinates logistics to delivery materials on site, reducing shopping time and costs in compliance with export law and regulations
- ensures subcontracts management
- provides relevant input for Project schedule and project physical progress
- identifies and supports the management of Supply Chain and Procurement risks and opportunities
- supports the Project Engineer in maintenance & management of the obsolescence register

4.3.3.3 Construction Manager

Main Responsibilities: The **Construction Manager** is responsible for planning, coordinating, and controlling all construction and installation activities during project execution. The role ensures that installation work (performed by external providers or internal resources) is delivered safely, on schedule, within scope and budget, and in compliance with quality and HSE requirements. The **Construction Manager [SRS]** monitors construction progress, manages construction interfaces and subcontractor performance, and ensures that all installation deliverables required for commissioning and project handover are accurate, complete, and timely.

Key Tasks

- defines and updates throughout the project, the construction installation strategy (including subcontracted activities) and relevant procurement technical specifications, supporting the process for procurement of installation work
- delivers people, plant, tools and test equipment necessary for construction activities
- performs site surveys prior to construction/installation activities and reports findings to Project Management
- defines, approves, and monitors the project construction schedule and provides physical progress updates to the Project Planner
- manages and coordinates, through the support of the site team, the external providers' installation activities, ensuring that all work is executed in compliance with the requirements, interface constraints, and the project execution plan

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- supports the Project Manager in forecasting installation costs, monitoring and issuing Installation Progress Payment Reports/Certificates for work performed by external providers
- exchanges technical information, designs, and manuals with partners/subcontractors to obtain method statements; ensure that a Method Statement for each installation activity is in place and reviewed with HSE support
- manages the amount of material installed respect the material delivery plan provided and updated by the SCPM
- monitors and ensures the availability of required construction/installation design documentation, method statements, materials, and adequate resources prior to installation
- ensures the Management of changes to installation works arising from or related to project design modifications
- ensures compliance with quality requirements for installation work
- ensures the implementation of HSE plan/rules on site during the installation phase
- ensures the handover to the Commissioning Manager upon completion of installation activities by formalizing the Handover Certificates and providing the relevant Installation Test Records
- captures and transfers the "as-installed" asset list as part of the data continuity process
- provides "as built" information to the Project Engineer

4.3.3.4 Commissioning Manager [SRS]

Main Responsibilities: Accountable for all commissioning activities on site

Key Tasks

- ensures the definition of the commissioning strategy and of the commissioning activities, the estimation of costs and resources, and the definition of the schedule; all regarding the on-site testing activities, including assumptions and limitations as well as commissioning coordination effort
- ensures the definition, execution and monitoring of commissioning activities, in accordance with the Commissioning Plan and/or Verification and Validation Plan and their integration and synchronization with the construction activities and the project delivery work plan and schedule. Ensures the reporting of physical progress
- ensures the on-time availability of resources, materials, tools and test equipment necessary for commissioning activities.
- ensures proper interface with the stakeholders (Customer Onsite Referents, Operators, Construction Managers and Commissioning Managers of partners/subcontractors, HSE Managers, V&V and RAMS for onsite testing

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activities, etc.) during the preparation and the execution of the commissioning and on-site testing activities

- ensures the Implementation of the established HSE plan/rules during commissioning
- ensures raising and management the non-conformity and change request with Project Engineer
- ensures the identification and management of risks and opportunities within commissioning

4.3.3.5 Warranty Manager [SRS]

Main Responsibilities: The **Warranty Manager** is responsible for the design of the Warranty activities, for people and resources mobilisation and for executing warranty activities according to the contractual requirements foreseen during the warranty period as well as managing the relevant financials

Key Tasks:

- ensures the Management of warranty teams operating at customer site or remotely
- with the support of the Supply Chain Planner and based on RAM data, ensures the managements of materials necessary for the execution of the warranty activities
- delivers people, plant, tools and test equipment necessary to delivery warranty activities.
- ensures the Monitoring of supplier performance (KPIs) in the execution of relevant warranty contractual obligations
- ensures the Collection of RAM data and keeps alive the FRACAS (Failure Reporting and Correcting Actions System) process hand overed to the Warranty team. Ensures proper reporting of the relevant information to relevant teams.
- ensures the Management of system updates, obsolescence of installed items and customer spares during the warranty period in conjunction with the Project Engineer and Procurement.
- ensures monitoring, control and reporting on the cost associated to warranty activities
- assesses requests for extended warranty or scope changes; price and formalise variations.

4.3.4 Project Execution Team – Roles applicable to VHC

4.3.4.1 Project RAM Manager

Main Responsibilities: The **Project RAM Manager** is responsible for the identification of the RAM requirements of the project, mapping applicability across the system/ component/ assembly, and ensuring the delivered solution is compliant.

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Key Tasks:

- analyses and defines RAM and LCC requirements based on contract, laws, regulations, or any other sources
- ensures the elaboration, review and update of RAM and LCC documentation
- supports design WPL in confirming the proposed design solution and managing any design changes
- develops RAM and LCC analysis for make subsystem
- coordinates sub-suppliers RAM and LCC activities
- integrates the RAM and LCC at vehicle level
- proposes, monitors and follows up remedial actions
- ensures the system/vehicle is compliant with RAM requirements prior to operations handover, including preparation of the RAM demonstration plan
- supports the Warranty manager preparing the RAM and LCC demonstration plan and in the analysis of data from service
- ensures elaboration of the first release of the maintenance plan and of its continuous updating
- identifies and supports the management of RAM and LCC risks and opportunities
- coordinates with Certification Manager regarding RAM certification process

4.3.4.2 Project Safety Manager

Tailoring or Adjustments: Reporting to the PM, the **Project Safety Manager** acts as a validator operating independently of the Project Engineer, in accordance with CENELEC/IEC standards

Main Responsibilities: The **Project Safety Manager** is responsible for the identification of the Safety requirements of the project, mapping applicability across the system/ component/ assembly, and ensuring the delivered designed solution is safe for the intended scope.

Key Tasks:

- analyses and defines safety requirements based on contract, laws, regulations, or any other sources
- ensures the elaboration, review and update of safety documentation
- supports design WPL in confirming the proposed design solution and managing any design changes
- coordinates sub-suppliers Safety activities
- monitors and follows up remedial actions
- identifies and, when required, manages and ensures the Safety Related Application Conditions (SRAC) are met/implemented

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- ensures the system/vehicle is compliant with Safety requirements prior to operations handover
- coordinates with Certification Manager regarding Safety certification process

4.3.4.3 Project Procurement Manager

Main Responsibilities: The **Project Procurement Manager (PPM)** manages and coordinates all the procurement activities relevant to the Project, ensuring the respect of budget, schedule, and quality of the deliverables

Key Tasks:

- analyses contractual requirements with the support of stakeholders (in particular Legal, Contract Management, and Project Management), ensuring key conditions are passed on to sub-suppliers
- issues the project procurement plan aligned with budget and schedule, and monitors progress, ensuring adherence
- completes of the supplier selection process
- tracks the completion of the First Article Inspection process
- ensures that all contracts/LOI relevant to key sub-suppliers are formally approved by the Project Manager
- verifies and monitors trend and deviations of procurement costs, supporting assessment of main causes and recover actions
- in case of change to vehicle configuration impacting supplier scope of work, analyses the impact of changes
- identifies and supports the management of Procurement risks and opportunities
- in case of make/buy mix changes, supports the Project Controller to estimate economic impact

4.3.4.4 Test Manager

Tailoring or Adjustments: In case of multi business project, **Test Manager** works in strict coordination with SRS Commissioning Manager, specifically for the performance of on-shore Type Test and System Integration Tests

Main Responsibilities: The **Test Manager** plans, coordinates and carries out testing activities for all vehicles used in development and verification activities (prototypes, pre-series), for the purposes of project validation, qualification and approval test, in accordance with the project test plan

Key Tasks:

- ensures that a Project Test Plan is developed and elaborates technical validation documentation on the basis of documentation issued by the PE

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- coordinates the elaboration of the vehicle testing procedures for routine and vehicle type tests, in accordance with the master schedule, ensuring the standardization of formats and processes for each plant involved
- ensures with PE that the procedures guarantee the full coverage required to validate homologation and design requirements at vehicle level
- supports the relevant WPLs for analysing vehicle test results
- support the Project Planner to ensure testing fully aligned with the second level schedule
- coordinates and agrees with project team / customer to ensure availability of required assets to perform any defined test activities
- defines and manages measurement systems and test tools for the presentation of test data
- provides feedback to the PE of the tests performed and the list of open items detected
- issues reports of tests carried out, in order to validate the requirements
- ensures the application of Health and Safety rules during the test phase
- cooperates with Design engineering units (who provide the Project Test Specification) and Integrated Planning in scheduling the Routine Testing of series build make components and vehicles in line with the entire operation planning
- works with the PE to support the Certification Manager, who is responsible for the vehicle authorization strategy
- assesses the impact (contractual, technical and economical) on activities of the operational railway rules of the countries in which Type Testing and the acceptance test have to be carried out

4.3.4.5 Project Operations Manager

Tailoring or Adjustments: Project Operations Manager (POM) for RS project is the former Project Manufacturing Manager. POM for S&M project is the former S&M Supply Chain Manager

Main Responsibilities:

Rolling Stock: coordinate and ensure the relevant industrial activities

S&M: coordinate and ensure the relevant maintenance activities

Key Tasks:

Rolling Stock (each plant involved in the RS project shall nominate a POM, as the interface towards the Project Team for industrialization, logistics and production activities of the relevant factory, ensuring the respect of production schedule, costs and quality)

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- reports progress of production activities for the relevant factory to the Project team on a regular basis (monthly progress reviews) and immediately in case of criticalities.
- supports the definition of the most competitive solutions (including technical solution, make or buy decision) providing all relevant elements related to manufacturing; feeding in the required manufacturing milestones that will be included in an Integrated Plan as an output measure for other Functions;
- ensures, by interfacing with the Planning of the Plant, the development and the regular update of the "Third Level schedule" of the plant manufacturing activities, in alignment with the overall schedule of the project;
- monitors the progress of the activities from the initial phases (including opening of the project, issue of drawings) to anticipate any impact on production activities and proposes recovery actions;
- organises the program of introduction of modifications and retrofits in the factory (including availability of materials, availability of production lines);
- coordinates the activities to transfer the vehicles/components among different factories;
- ensures the readiness of the vehicle/component for its delivery to the customer; in particular is responsible for the vehicle/component configuration (changes, missing parts, nonconformities) and for related update.
- in case of more plants manufacturing the same Scope of Work, a coordination among POMs is necessary during the industrialization and change management phases

Service and Maintenance

- implements and governs the contracted service model and maintenance strategy in compliance with client requirements
- plans and optimises resources, materials and operational capacity to meet availability and reliability targets
- monitors maintenance activities and contractual KPIs (availability, reliability, maintainability), anticipating issues and defining recovery actions
- coordinates logistics activities, ensuring regulatory compliance and optimisation of lead times and costs
- drives continuous improvement of maintenance processes, including optimisation of maintenance intervals and introduction of diagnostic solutions

4.3.4.6 Commissioning Manager [VHC]

Main Responsibilities: Leads and controls all on-site commissioning activities required to deliver vehicles to the customer, ensuring safe execution and achievement of test, repair and acceptance milestones within the agreed budget and schedule

Key Tasks:

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- plans and coordinates all commissioning, repair and completion activities at customer premises, ensuring optimal train configuration and positive testing prior to the delivery to the customer
- sizes and manages of commissioning staff workload
- manages of materials flow and interfacing with suppliers for interventions
- interfaces with the customer for test activities undertaken on external sites
- implements the established HSE plan/rules during commissioning
- identifies and supports the management of risks and opportunities associated with the Commissioning period

4.3.4.7 Warranty Manager [VHC]

Main Responsibilities: The **Warranty Manager** is responsible for the design of the maintenance activities, people and resources mobilisation and for executing maintenance activities according to the contractual requirements foreseen during the warranty period

Key Tasks:

- manages of maintenance teams operating at customer site or remotely
- with the support of RAM data, manages materials necessary for the execution of the warranty
- monitors supplier performance in the execution of relevant warranty obligations
- collects of RAM data
- where applicable, supports the execution of system updates to be carried out at customer site or from remote
- identifies and supports the management of risks and opportunities associated with the Warranty period
- implements the established HSE plan/rules

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5 TRANSITION RULES - SUMMARY

RULE #	DOMAIN	APPLICABLE DOC	SUPERSEDED DOCS (*)	PERIOD
1	Description: Bid or Project Team Roles Rationale: Full harmonisation of Bidding and Project Team roles still in progress.	RSBU Legacy Document: G-MNL A0800(**) GTS Legacy Document: 5.2 PRJ – Manage Project 87200984-00-PRJ-GRP-EN(**) 5.1 BID - Manage Bid 87200484-00-BID-GRP-EN(**)	Description: N/A	Until the release of the full list of Bidding and Project Team roles and their associated tailoring

(*) The corresponding MS Administrator is accountable for removing the superseded documents from the MS repository.

(**) Partially superseded by transition rule §4 excepted for roles not under the initial release of the PTO or additional detail related to tailoring for specific projects.

Although this document is applicable immediately upon publication, it is understood the RSBU have a large backlog of complex Bids and Projects in execution and will be unable to transition to the new Organisation instantaneously.

For all new Bids or Projects beginning execution after the date of publication, the roles within this TRN will be applied. These roles may be supplemented by roles in documentation from legacy RSBU or legacy GTS only where these are not covered in the transition rules and they do not conflict with any principles defined within this document. Upon the release of the full Project Team Organisation Manual, legacy documentation will be fully superseded.

For Bids currently in progress, the roles defined in this document shall not be applied if the bid is expected to be completed within three (3) months from the date of publication of this document. For Bids with an expected completion beyond this period, the applicability of the new roles and organisation shall be assessed on a case-by-case basis, in agreement with the respective LoB/Region/BS management.

For Projects currently in execution, it is expected that a proportion of the backlog projects will be required to transfer to the new Project Team Organisation. This will follow a staged approach defined by the PMO Organisation in accordance with LoB/region/BS management, based on the value of the project, percentage of completion, the remaining time to completion, its structure (extent of multi-site/segment/LoB collaboration), and risks or opportunities perceived by the Project Management organisation.

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6 COMPARISON SUMMARY

A comparison between the former RSBU and legacy GTS Bidding and Project Team Organisations has been performed. Following this analysis, the following main points of difference and similarity are noted:

- Both legacy businesses operate in a project environment and have a number of established standard roles that are staffed in case of Bidding or Project execution.
- Both legacy businesses require collaboration between a Bidding or Project organisation and a function organisation in order to meet the objectives of the business.
- Differences exist in the legacy organisations in the specific roles required as standard in a Bidding or Project Team (roles, naming, responsibilities and key tasks), e.g. legacy RSBU Sales Manager and legacy GTS Capture Lead.
- Differences also exist in the management of Bidding or Project Team roles when a project operates across multiple regions, business segments, or Lines of Business, e.g. a single accountable PM in legacy RSBU and multiple PMs in legacy GTS.
- Definition of a Bidding or Project Team Organisation is managed with different levels of formality between the two legacy organisations, with legacy RSBU using a GBMS Manual and legacy GTS leveraging an excel tool explaining roles.

The results of the comparison analysis, including identified best practices on both sides, have been taken into consideration in defining and clarifying the principles and Bidding and Project Team roles defined in this document.

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